

Fig. 6: Design a filter by selecting the elements and entering their values

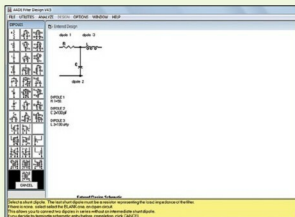


Fig. 7: Entered schematic, the list of dipoles used, and their values

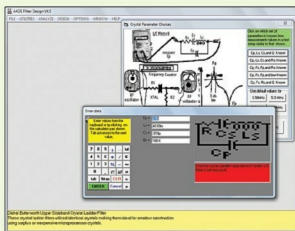


Fig. 8: Creating a Disha Butterworth USB crystal ladder filter

can get through the filter.

Order. The order of the filter.

Note. Either the stopband attenuation or the order of the filter can be specified, but not both.

Input impedance (Z_{in}) and Output impedance (Z_{out}). Enter the impedances at either end of the filter. These only affect the values of the two resis-

tors in the design.

Click Enter. In the resulting window, enter the value of the unloaded Q (Q_u) of the inductor, if known. Else, leave it as it is and the inductor would be assumed ideal. Enter a title for this design and click OK. Fig. 4 shows the resulting window where the circuit and component values will now appear.

One can choose the response to plot for a given design from *Analyze* on the menu bar. Here is an example of how power insertion gain is plotted.

Go to *Analyze* > *Power insertion gain*. Enter the requested plotting parameters—the range of frequencies (Freq1 and Freq2) and the range of attenuation (db1 and db2) to be plotted. Click *Enter* to view the plot. The power insertion gain response of a fifth-order Chebyshev low-pass filter with default values is shown in Fig. 5.

Creating a new schematic

The AADE Filter Design allows users to create their own computer schematic from a pre-existing design and analyse its response.

You can reconstruct the design from a book or magazine into the program's workspace using the dipole blocks given.

Go to *Utilities* > *Enter Design*. A plain workspace, *Entered Design*, would now appear. The elements for designing will be shown on the left of the workspace, under the heading *Dipoles*.

Begin the design by selecting the first element, a series resistance, which represents the source impedance of the filter. Clicking on the icon representing each dipole opens up a window in which you can change the element's configuration or value (refer to Fig. 6). You can also use a transformer to change the impedance of dipoles that follow.

As you insert each dipole, these automatically appear as the circuit in the main workspace, as shown in Fig. 7. The list of dipoles and their values used will be displayed just below the circuit. This can be disabled by unchecking *Options* > *Parts List*.

The last shunt dipole in the design should be a resistance, which represents the load impedance. If it is an open circuit, click on the blank icon in *Dipoles*. The blank icon also allows users to connect two dipoles in a series without a shunt dipole in between.

Other functions

As mentioned before, you can either prepend (insert before) or append other filters to the current filter in the workspace. For that, you first need to have a filter file (in *.sch format) already saved in the computer.

Now go to *Utilities* > *Append Filter/Insert Filter*. Select the file you want to append/prepend. That design will be conjoined with the current design to create a new filter.

This is mainly used to design broadband band-pass filters.

A great tool

AADE Filter Design is a great Windows tool for designers, hobbyists and students to design and analyse crystal and LC filters. You just have to input the filter parameters, and AADE Filter Design will generate the filter, provide a list of optimum values of the components, and let you plot response curves. Its simplicity and extensive help guide makes it one of the best freeware available on the Internet for passive filter design. ●

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